

How effective is Black Hole Algorithm?

Himanshu Gupta¹, Arpit Gupta², Shubham Kumar Gupta³, Pranshu Nayak⁴, Tanmay Shrivastava⁵

³Electrical Engineering Dept. , MITS ,Gwalior

^{1,2}Electronics & Telecommunication Dept. , SGSITS ,Indore

⁴Electronics & Telecommunication Dept. , LNCET (Formerly GGITM) Bhopal

⁵Electrical Engineering Dept. , SGSITS ,Indore

Abstract— Metaheuristics have become popular in solving optimization problems. Recently literature has been flooded with lot of “novel” optimization techniques. These techniques are inspired by various natural phenomena. One such technique is Black Hole Algorithm, which is inspired by the Black Holes. The author of this technique claim it to be better than Particle Swarm Optimization(PSO) ,but we have found it contrary. In this paper we compare the Black Hole Algorithm and Particle Swarm Optimization(PSO) by evaluating them on standard test suite. The results show that BHA performs very poorly as compared to PSO and thus ,falsifying the claim made by authors of BHA.

Keywords—Particle Swarm Optimization, Black Hole Algorithm , Metaheuristic

I. INTRODUCTION

Practical optimization problems are characterized by a computationally expensive objective function with large number of variables and constraints. These problem are difficult to tackle with calculus and gradient based methods and call for more sophisticated numerical techniques. Metaheuristics are one of the numerical techniques which have become increasingly popular for solving optimization problems. Genetic Algorithms[1], Differential Evolution[2], Simulated Annealing[3], Tabu Search[4], Particle Swarm Optimization[5] are some of the popular metaheuristic algorithm. The field of metaheuristic is relatively new and a lot of research work is being carried out. Recently, the literature has seen a tremendous increase in new metaheuristic algorithms. These metaheuristic algorithms have drawn inspiration from various natural sources like Wolf Pack[6] , Krill Herds[7], Weeds [8] , Moth flies[9] , Ions [10] , Gravity[11] ,Black Holes[12] etc. Many of these algorithms have proved to be effective and powerful, but some of them are not as efficient as they claim to be. Black Hole Algorithm (BHA)[12] is one of such metaheuristic that had made big claim for it's effectiveness[13] , but in reality it's performance is poor. In this paper, we study the performance of Black Hole Algorithm and compare it with Particle Swarm Optimization. The tests are carried out on standard test suite [14], which consists of 25 optimization problems and results are examined.

The paper is organized as follows; in section II and III we discuss briefly BHA and PSO. In section IV the test setup for evaluation is introduced , followed by results in section V .Finally the paper is concluded in section VI.

II. BLACK HOLE ALGORITHM

Black Hole Algorithm[12] is nature inspired, population based metaheuristic which is based on the behavior of black holes. A black hole is a region of very strong gravitational pull. The gravitation pull is so strong that even light cannot escape from it. The region from which no escape is possible is called event horizon.

In Black Hole Algorithm, an initial population of candidate solutions called stars is generated. The star with best fitness is selected and made Black Hole. Each star then moves toward the Black Hole. The movement of star is governed by following equation:

$$X^k(t+1) = X^k(t) + \text{rand} * (X_{BH} - X^k(t)) \quad (1)$$

Where

$X^k(t+1)$ = Position of k^{th} Star at $(t+1)^{\text{th}}$ iteration

$X^k(t)$ = Position of k^{th} Star at t^{th} iteration

X_{BH} = Position of Black Hole

When star moves toward black hole it can move to a point where the fitness is greater than of the black hole. In that case, the black hole is shifted to the new position. There can also a case when a star crosses the event horizon, in such case, the star is absorbed by black hole and new star is generated randomly in search space. The radius of event horizon is given by equation (2):

$$R = \frac{f_{BH}}{\sum f_i} \quad (2)$$

Where

R = radius of event horizon

f_{BH} = fitness of black hole

f_i = fitness of i^{th} star

Steps for BHA Algorithm:

1) Initialize the stars randomly in search space.

Loop

- 2) Evaluate Objective Function for each star
- 3) Select star with best fitness and make it blackhole
- 4) Move the stars towards blackhole according to equation 2
- 5) If star reaches to a location where fitness is better than blackhole, exchange their cities.
- 6) If star crosses event horizon, initialize it with random position again
- 7) If the termination condition is met, exit the loop.

End of Loop

III. PARTICLE SWARM OPTIMIZATION

Particle Swarm Optimization is metaheuristic introduced by Eberhart and Kennedy [5] which is inspired by the flocking behavior of birds. PSO begins with set of candidate solutions (particles). These particles fly in multi dimensional search space of objective function finding the optimum solution. The velocity with which particles move in search space is updated every generation and is given by:

$$V^k(t+1) = V^k(t) + c1 * r1 * (pbest - x^k(t)) + c2 * r2 * (gbest - x^k(t)) \quad (3)$$

Here,

$V^k(t)$ = velocity of particle at instant t

$gbest$ = it is best position encountered by whole swarm

$pbest$ = it is best position encountered by a particular particle

$c1$ = social parameter

$c2$ = cognitive parameter

$r1$ and $r2$ = random number uniformly sampled from [0,1]

The parameter $c1$ and $c2$ decide how much a particle's velocity is affected by its past positions and the neighbourhood. The position of particle in search space is then given as:

$$X^k(t+1) = X^k(t) + V^k(t) \quad (4)$$

Here,

$X^k(t)$ = position of particle at time t

The main steps for PSO are outlined below:

Steps for PSO Algorithm:

- Initialize the particles position and velocities
- Evaluate objective function and find $gbest$ and $pbest$

Loop

- Update the velocity as per equation
- Update the position as per equation
- Update $gbest$ and $pbest$
- If terminal condition is met, break for loop else continue

End of Loop

IV. TEST SETUP

The comparison tests are conducted using Test Suite of CEC 2005 as described in [14]. There are overall 25 test problem out of which 5 are unimodal problems, and 20 are multimodal optimization problems. The basic description of problem is given in table below:

Problem	Type	Optimum Value	Range
F1	Unimodal	-450	$[-100,100]^D$
F2	Unimodal	-450	$[-100,100]^D$
F3	Unimodal	-450	$[-100,100]^D$
F4	Unimodal	-450	$[-100,100]^D$
F5	Unimodal	-310	$[-100,100]^D$
F6	Basic Multimodal	390	$[-100,100]^D$
F7	Basic Multimodal	-180	$[0,100]^D$
F8	Basic Multimodal	-140	$[-32,32]^D$
F9	Basic Multimodal	-330	$[-5,5]^D$
F10	Basic Multimodal	-330	$[-5,5]^D$
F11	Basic Multimodal	90	$[-0.5,0.5]^D$
F12	Basic Multimodal	-460	$[-\pi,\pi]^D$
F13	Expanded Multimodal	-130	$[-3,1]^D$
F14	Expanded Multimodal	-300	$[-100,100]^D$
F15	Hybrid Composition	-120	$[-5,5]^D$
F16	Hybrid Composition	-120	$[-5,5]^D$
F17	Hybrid Composition	-120	$[-5,5]^D$
F18	Hybrid Composition	10	$[-5,5]^D$
F19	Hybrid Composition	10	$[-5,5]^D$
F20	Hybrid Composition	10	$[-5,5]^D$
F21	Hybrid Composition	360	$[-5,5]^D$
F22	Hybrid Composition	360	$[-5,5]^D$
F23	Hybrid Composition	360	$[-5,5]^D$
F24	Hybrid Composition	260	$[-5,5]^D$
F25	Hybrid Composition	260	$[2,5]^D$

Each algorithm is run for 30 times on each problem for 10,30 and 50 dimensions. Maximum iterations is 2000 for each algorithm. The swarm size of PSO and stars population for BHA both is fixed to 50. Value of c1 and c2 for PSO is 2.05 and 2.05 respectively.

V. RESULTS

The results are tabulated in table 2, for 10 dimensions, table 3, for 30 dimensions, table 4, for 50 dimensions. From the results it is clear that PSO outperforms Black Hole Algorithm on every problem. On problems 8 and 11, however the results of BHA are comparable to that of PSO. An interesting fact to note is that the problems 8 and 11 are both multimodal problems. But, overall PSO performs better than BHA.

One may argue as per the No Free Lunch (NFL) Theorem [15], that BHA may outperform PSO on some other problem set; but in practical pre-conditions of NFL lunch theorems are no met, and NFL theorem don't hold [16].Also ,the problem set where BHA will outperform PSO might not be of interest in practical. Hence in practical , we can expect some algorithms to perform better than others. However, there can still be some cases in simple algorithm to outperform other complex ones.

The results make the picture clear and show that PSO is much better than BHA, contrary to the claims made by authors in [13].

Table1.Results for 10 Dimensions

10	<i>PSO</i>			<i>BHA</i>		
Probl em	Mea n	Medi an	Std. Dev	Mea n	Medi an	Std. Dev
F1	-450	-450	1.3e-09	1.03e+04	9.70e+03	4.05e+03
F2	-450	-450	0.0053	1.408e+04	1.421e+04	5.221e+03
F3	6.946e+04	1.016e+05	1.168e+05	1.886e+06	2.007e+06	9.171e+05
F4	- 448.1 5756 1	- 449.8 0572	3.2097	1.6879e+04	1.5366e+04	5.1615e+04
F5	1547.933	1504.7214	338.0085	1.8232e+04	1.8277e+04	1919.5239
F6	408.5572	395.7531	37.99	2.1677e+09	1.1914e+09	2.1686e+09
F7	- 179.8 224	- 179.9 743	2.3676	- 178.8 186	- 178.8 104	1.072
F8	- 119.6 287	- 119.6 419	0.1299	- 119.3 327	- 119.2 01	0.20018

F9	- 327.5 466	- 327.8 962	3.340	1261. 7635	1105. 4833	758.68 44
F10	- 316.7 789	- 318.0 617	4.8528	5384. 7221	5857. 8193	3084.2 694
F11	96.870	97.5713	1.007	101.4571	102.167	1.7316
F12	1.7502e+04	1.5893e+04	1.0592e+04	9.4621e+04	9.7774e+04	3.6186e+04
F13	- 129.1 515	- 129.1 728	0.2596	1.6165e+11	2.8521e+11	3.2034e+11
F14	- 296.6 995	- 296.5 142	0.3482	295.7318	295.7730	0.1451
F15	460.6875	544.487	151.2464	480.9236	514.7667	137.7903
F16	251.7121	248.0678	14.0773	320.0056	320.6547	35.3201
F17	261.833	261.2879	19.5337	340.1676	342.4567	38.0907
F18	608.4676	560.00381	316.9709	2.0078e+03	2.0131e+03	17.7006
F19	799.4247	921.0465	249.0950	2.0246e+03	2.0216e+03	9.2754
F20	657.1937	810.0502	280.1310	2.0175e+03	2.0214e+03	18.6118
F21	820.0009	860.0	84.3274	3.0228e+03	2.8907e+03	497.4785
F22	1161.7534	1152.6966	40.6573	2.1574e+03	2.1973e+03	256.592
F23	1126.3676	1081.2222	126.1442	2.9548e+03	2.8811e+03	338.1253
F24	590.2213	460	235.9378	1.608e+03	1.5216e+03	298.327
F25	1043.2247	998.7621	240.3767	2.3004e+03	2.3087.885e+03	214.239

Table 3. Results for 30 Dimensions

30	<i>PSO</i>			<i>BHA</i>		
Proble m	Mean	Media n	Std. Dev	Mean	Meda in	Std. Dev
F1	- 443.08 02	- 445.98 72	9.506 2	6.685 4e+0 4	6.507 6e+0 4	8642. 1915
F2	5837.4 7447	5906.4 6411	2450. 7724	8.349 8e+0 4	7.301 2e+0 4	2.638 6e+0 4
F3	4.788e +06	6.450e +06	6.145 e+06	8.404 e+07	8.310 e+07	9.863 e+06
F4	1.7031 e+04	1.6451 e+04	6377. 6385	6.731 6e+0 4	6.494 3e+0 4	1.086 7e+0 4
F5	6316.6 459	6372.8 673	1032. 7041	4.290 2e+0 4	4.236 5e+0 4	4.661 9e+0 3
F6	3061.0 27	701.55 62	4005. 21	3.389 6e+1 0	3.607 1e+1 0	1.051 6e+1 0
F7	2760.9 4135	2751.4 3561	26.91 94	9.176 1e+0 4	8.971 6e+0 4	2.141 e+04
F8	- 118.96 12	- 118.94 54	0.081 6	- 118.8 277	- 118.8 2	0.085 29
F9	- 283.33 9	- 281.45 83	12.20 12	1.367 5e+0 4	1.331 5e+0 4	5049. 103
F10	- 234.82 21	- 246.26 493	32.64	2.488 9e+0 4	2.459 3e+0 4	5166. 161
F11	131.57 81	131.55 77	0.985 9	134.8 231	134.7 865	2.753 71
F12	9.3538 e+05	9.2313 e+05	2.355 9e+0 5	1.772 7e+0 6	1.668 2e+0 6	3.450 1e+0 5
F13	- 120.58 02	- 120.54 66	3.082 2	1.607 2e+1 4	9.516 1e+1 3	2.147 2e+1 4
F14	- 286.58 53	- 286.68 39	0.236 4	- 285.8 7814	- 285.7 87	0.241 53
F15	433.00 94	444.11 63	117.1 375	86.87 24	88.83 671	71.45 24
F16	460.24 39	446.33 35	74.14 21	70.36 71	70.66 73	7.024 2
F17	469.65 88	495.10 07	74.58 19	74.18 29	75.12 43	9.395 9

F18	952.07 53	952.31 148	6.424	2077. 526	2077. 7878	11.28 211
F19	947.33 32	947.93 93	3.946 20	2074. 287	2073. 7843	4.407 53
F20	947.71 63	947.12 87	2.698 4	2079. 054	2074. 0919	23.59 91
F21	891.78 13	860.57 93	98.52 24	7941. 5429	1.183 4e+0 4	1.617 9e+0 4
F22	1352.0 185	1354.6 364	16.92 166	4.045 6e+0 4	3227. 5102	8.112 0e+0 4
F23	961.64 64	899.22 74	199.4 967	1.708 2e+0 4	5424. 938	2.560 3e+0 4
F24	460.58 65	460.62 01	0.237 2	3138. 7676	2253. 4923	2086. 992
F25	490.15 27	490.12 21	9.353 2	596.3 451	939.4 798	491.5 47

Table 4. Results for 50 Dimensions

50	<i>PSO</i>			<i>BHA</i>		
Proble m	Mean	Medi an	Std. Dev	Mean	Meda in	Std. Dev
F1	- 438.1 454	- 440.2 132	14.22 3718	1.288 7e+05	1.273 77e+0 5	1.333 8e+04
F2	4.496 1e+04	4.211 2e+04	1.310 1e+04	3.997 4 e+05	3.746 9e+05	1.418 3e+05
F3	3.528 e+07	3.864 e+07	1.951 e+07	7.658 e+08	7.694 e+08	9.071 e+07
F4	8.504 6e+04	9.108 6e+04	1.829 6e+04	1.301 0e+05	1.295 8e+05	1.023 8e+04
F5	1.584 7e+04	1.455 7e+04	3.102 e+03	4.548 2e+04	4.457 2e+04	4.661 9e+03
F6	3.535 e+05	2.657 2e+05	3.037 4e+05	7.528 6e+10	7.890 8e+10	1.510 9e+10
F7	3869. 4024	3873. 5477	23.13 250	3.573 6e+05	2.767 1e+05	2.537 9e+05
F8	- 118.7 795	- 118.7 942	0.065 4	- 118.6 681	- 118.6 57	0.036 5
F9	- 210.6 354	- 211.8 162	30.40 8	2.468 4e+04	2.409 5e+04	2166. 670
F10	- 140.7 659	- 164.9 549	80.94	5.147 1e+04	5.333 7e+04	3569. 358

F11	165.4 358	164.9 086	2.714 7	168.5 465	168.9 233	2.714 7
F12	4.404 7e+06	4.196 7e+06	1.050 7e+06	8.468 1e+06	8.351 e+06	1.050 7e+06
F13	- 100.3 063	- 100.0 425	4.292 4	2.672 1e+14	2.376 4e+14	1.724 e+14
F14	- 276.6 368	- 276.6 305	0.231 142	276.1 469	276.0 748	0.385 8
F15	523.5 282	525.0 309	90.44 42	949.8 325	1003. 0146	107.2 064
F16	466.9 957	449.4 911	52.01 44	860.8 456	861.7 271	58.10 4
F17	504.7 299	487.4 422	87.68 65	919.3 473	916.8 992	58.90 64
F18	993.3 911	994.4 351	8.129 7	2119. 54788	2081. 7432	88.65 26
F19	996.2 416	991.6 039	13.23 55	2105. 5934	2097. 97	37.64 71
F20	996.1 203	994.9 045	9.919 2	2154. 3182	2092. 5478	112.6 945
F21	1375. 1914	1467. 8940	200.1 050	3.768 1e+04	1.220 7e+04	4.291 0e+04
F22	1436. 74612	1428. 6820	40.95 310	4.891 e+04	6.495 5e+04	1.326 6e+06
F23	1433. 82101	1468. 3934	72.88 096	5.096 3e+04	2.241 1e+04	5.793 1e+04
F24	556.2 298	527.3 6626	62.42 8	3.403 6e+04	9.407 e+03	4.520 2e+04
F25	602.7 213	851.3 02	436.3 892	1.660 e+04	1.666 e+04	4.890 E+02

VI. CONCLUSION

In this paper we have compared both PSO and BHA on 10,30 and 50 dimensional problems. It is seen in results that BHA performs very poorly as compared to PSO. However, on some problems BHA's performance is comparable to PSO. But we can say that overall PSO is way much better than BHA, and BHA is simply no match for PSO.

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